

# Governed Clinical Analytics Copilot

A self-hostable NL→SQL agent over de-identified EHR data, fine-tuned on OmniSQL-7B, achieving state-of-the-art performance on the EHRSQL 2024 benchmark with a fully open-source, 7-billion-parameter model.

| OUR RS(10)                      | VS. EHRSQL 2024 WINNER                      | DELTA                                 |
|---------------------------------|---------------------------------------------|---------------------------------------|
| 0.873                           | 0.813                                       | +7.4%                                 |
| Full test set · 1,167 questions | LG AI / KAIST · GPT-3.5 SFT + self-training | Open 7B model vs. proprietary GPT-3.5 |

## Project Overview

Clinical data analysts spend hours writing SQL against complex EHR schemas. This project builds a governed, self-hostable copilot that translates plain-English questions into verified SQL queries over the MIMIC-IV database — and critically, **knows when to abstain** rather than hallucinate an answer for an unanswerable question.

We benchmarked against **EHRSQL 2024**, a shared task from NAACL 2024 ClinicalNLP challenging models with 1,167 test questions: 934 answerable and 233 intentionally unanswerable. The RS(10) metric penalises hallucinations 10× more than missed abstentions — making abstention quality the dominant factor.

Using **OmniSQL-7B** (Qwen2.5-Coder-7B-Instruct pre-trained on SynSQL-2.5M) as the base, we designed a supervised fine-tuning pipeline that teaches the model the MIMIC-IV schema, correct abstention behaviour, and zero-shot clinical NL→SQL — all without proprietary APIs.

## Key Results

| Metric              | Value          | Detail                                                  |
|---------------------|----------------|---------------------------------------------------------|
| RS(10) Score        | 0.873          | Primary benchmark metric — full 1,167-question test set |
| Execution Accuracy  | 92.4%          | 863 / 934 answerable questions answered correctly       |
| Hallucinations      | 7 / 233        | Wrong SQL on unanswerable questions (3.0%)              |
| Correct Abstentions | 226 / 233      | Model correctly refused 97% of unanswerable questions   |
| Model Size          | 7B params      | Open weights, self-hostable, no API dependency          |
| Training Examples   | 53,006         | SFT with 3× unanswerable oversampling                   |
| Training Hardware   | 96 GB GPU      | Google Colab RTX 6000 Pro — 2 SFT epochs                |
| Local Eval (100-Q)  | RS(10) = 0.940 | EX=92.5%, 0/20 hallucinations, 20/20 abstentions        |

# Benchmark Comparison — EHRSQL 2024 MIMIC-IV

| System                             | Model                       | EX    | Halluc. | Abstain. | RS(10) |
|------------------------------------|-----------------------------|-------|---------|----------|--------|
| Ours (SFT-only) ★                  | OmniSQL-7B · open           | 92.4% | 7/233   | 226/233  | 0.873  |
| EHRSQL 2024 Winner (LG AI / KAIST) | GPT-3.5 SFT + self-training | ~75%  | —       | —        | 0.813  |
| EHRSQL 2024 3rd (ProbGate)         | GPT-3.5 + log-prob filter   | —     | —       | —        | 0.742  |
| OmniSQL-7B baseline                | No fine-tune                | 61.2% | 133/233 | 100/233  | -0.564 |

## What We Built — End-to-End Pipeline

### Phase 1: Data Engineering

Built 53,006 SFT training examples from EHRSQL train + valid + train\_aug splits. Oversampled unanswerable questions 3× to match the 20% test distribution (vs. 8.9% in training data). Annotated answerable questions with gold SQL; unanswerable with [ABSTAIN]. Output: data/sft/omnisql\_sft\_train.jsonl (184 MB).

### Phase 2: SFT Training — RTX 6000 Pro (96 GB)

Fine-tuned OmniSQL-7B with Unsloth + 4-bit quantisation. LoRA  $r=32$ ,  $\alpha=64$ , 7 target modules (q,k,v,o,gate,up,down). 2 epochs, effective batch size 16, cosine LR schedule at  $1e-4$ . Trained on the full 2,733-character MIMIC-IV schema-inclusive system prompt. Adapter size: 323 MB, 392 LoRA tensors.

### Phase 3: ORPO Preference Training

Built 492 preference pairs from baseline predictions (359 wrong-SQL + 133 hallucination pairs). Trained 1 additional epoch at  $lr=2e-6$  with ORPO  $\beta=0.1$ . Key finding: SFT-only (0.873) outperformed SFT+ORPO due to asymmetric gradient dynamics — [ABSTAIN] is 1 token vs SQL at 50–200 tokens.

### Phase 4: Evaluation Infrastructure

Built local evaluation harness (quick\_eval\_adapter.py) with stratified 80+20 sampling (SEED=42), batch inference, SQL execution against MIMIC-IV SQLite, and RS(N) scoring. Full harness supports self-consistency voting, entropy filtering, execution-guided repair, and RAG few-shot retrieval (BM25 + embed hybrid).

## Critical Technical Lessons

|                                            |                                                                                                                                                                                                                  |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Prompt Must Include the Full Schema | The full prompt is 2,733 chars including the entire MIMIC-IV schema. The fallback stub (548 chars) provides no schema — model abstained on 99% of everything. A single file fix moved RS(10) from 0.210 → 0.873. |
| SFT Eval Format Must Match Training Format | Model trained on bare questions in the user turn. Few-shot eval prepended Q&A; preambles the model had never seen. Zero-shot (K=0) is the correct eval format for this model.                                    |

|                                                           |                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ORPO Overcorrected Abstention</b>                      | [ABSTAIN] is 1 token (concentrated gradient) vs SQL at 50–200 tokens. With 27% abstention pairs, ORPO overcorrected — model abstained on 99%+ even when it should generate SQL. ORPO pairs must be built from the SFT model's own failures, not the baseline. |
| <b>Unanswerable Class Imbalance Is the Core Challenge</b> | Train: 8.9% unanswerable. Test: 20% unanswerable. Without 3× oversampling, the model never learns reliable abstention. Our strategy produced 97% correct abstention rate.                                                                                     |
| <b>Entropy Calibration Must Match Eval Format</b>         | Post-SFT model entropy maxed at 0.0401 — effectively zero variation. Calibration must be re-run after any prompt or format change. With only 7 remaining hallucinations, correct calibration could push RS(10) → 0.90+.                                       |
| <b>Open 7B Beats Proprietary GPT-3.5</b>                  | OmniSQL-7B's SynSQL-2.5M pre-training provides rich SQL patterns. Domain-specific SFT makes it exceed GPT-3.5 SFT + self-training at a fraction of the inference cost — fully on-premise with no API dependency.                                              |

## What's Next

| Next Step                   | Description                                                                                                                                                               | Potential Impact            |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Entropy Calibration</b>  | Re-calibrate abstention threshold with correct system prompt. Only 7 hallucinations remain — catching most of them could push RS(10) above 0.90.                          | <b>RS(10) → ~0.90+</b>      |
| <b>Self-Training Pass</b>   | Apply SFT model to 38K train_aug examples, collect high-confidence [ABSTAIN] pseudo-negatives, run 1 additional SFT epoch. Mirrors EHRSQL 2024 winner's Stage 2 strategy. | <b>RS(10) → ~0.91–0.93</b>  |
| <b>Corrected ORPO Pairs</b> | Build pairs from SFT model's own failures (not baseline). Cap abstention pairs at 10–15% of total. Use ORPO $\beta=0.05$ to reduce abstention gradient dominance.         | <b>+1–2% RS(10)</b>         |
| <b>Governed Deployment</b>  | Package adapter + harness as self-hosted API with audit logging, query provenance, and role-based access control.                                                         | <b>Production milestone</b> |